

## Java program

### Simple Java program

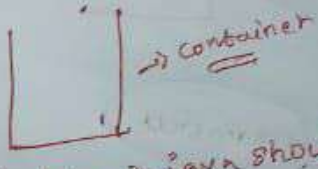
How to write a simple Java program

public class First

```
{  
    public static void main (String[] args)  
    {  
        System.out.println("Hello world");  
    }  
}
```

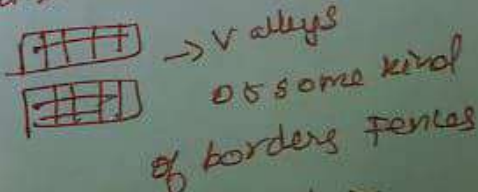
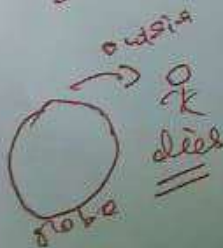
}

\* class →



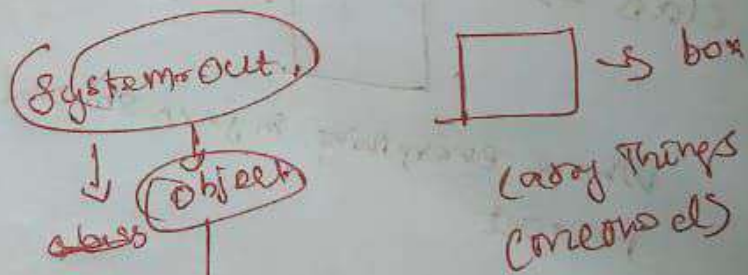
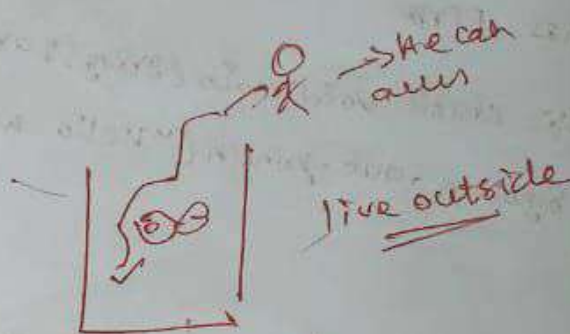
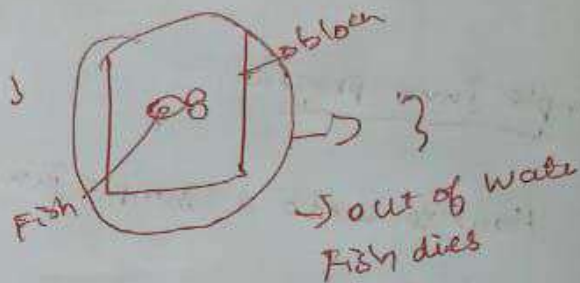
Where everything in Java should  
present

} → Think this like some kernel



If something starts here  
dies at }

2

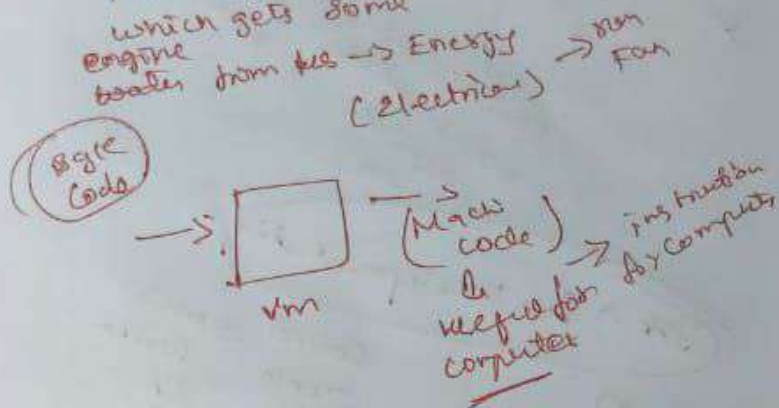


print(nl) → prints the  
Everything in terminal

# Java Virtual Machine



Think like a machine  
which gets some  
energy from the

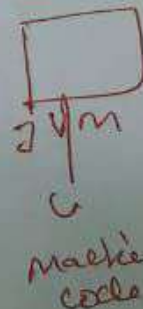


Java

• Java  
code



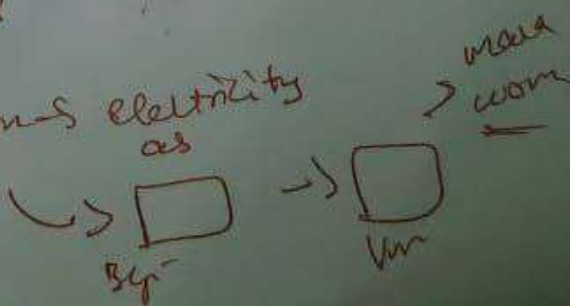
→ Byte Code



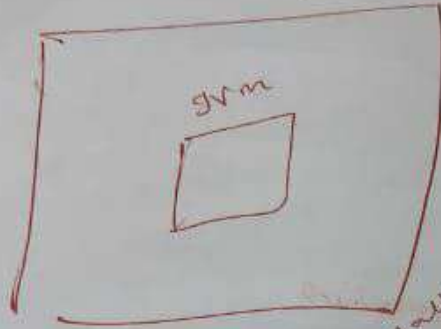
Like Exam

Electricity = load

↓  
some electricity as







Java Runtime Environment

Environment which makes machine run

Job → Java Development kit

Compiler Tools → make write do mfg

Like tools to maintain something in main



makes Tools more electr



electronic products

just JVM

Job  
↓  
JVM JRE

JVM  
↓  
comes first → main method

↳ designed to say a word a a child  
↳ Thinks something as mainly his/her mother

Public Accessible  
Accessibility

Public static void main (String args[])  
{  
    ↳ systems set-up  
}  
    ↳  
    ↳ object  
    ↳ port 1a

Comments

Comment  
    ↳ something you write  
    ↳ code age

You could forget

1) ↳ write line 1 → For own  
    ↳ other  
    ↳ understand

Not for computer

// -> single line

/\* \*/



Sometimes to  
stop code from  
running

/\* \*/ we don't use  
(rule) language  
Not nested

## Data Types



country  
persons → integers (numbers)  
→ int



name → string  
person



yes  
no  
something

→ true or false  
→ boolean



float

decimal scale

→ ., point, dot



int

|       |
|-------|
| byte  |
| short |
| int   |
| long  |

specific size to store numbers

|        |
|--------|
| float  |
| double |

|         |
|---------|
| boolean |
|---------|

|      |
|------|
| char |
|------|

char is 1 byte

Primitives

Primitives

I'm here to explain about memory



Like this memory you cannot see in your brain

|            |
|------------|
| stack      |
| primitives |

|      |
|------|
| heap |
|------|

|        |
|--------|
| main   |
| static |

|       |   |                 |                                  |
|-------|---|-----------------|----------------------------------|
| byte  | → | 1 byte 8 bits   | $-2^8 \text{ to } 2^8 - 1$       |
| short | → | 2 bytes 16 bits | $-2^{15} \text{ to } 2^{15} - 1$ |
| int   | → | 4 bytes 32 bits | $-2^{31} \text{ to } 2^{31} - 1$ |
| long  | → | 8 bytes 64 bits | $-2^{63} \text{ to } 2^{63} - 1$ |

int → most situation

12 → defau (cm)

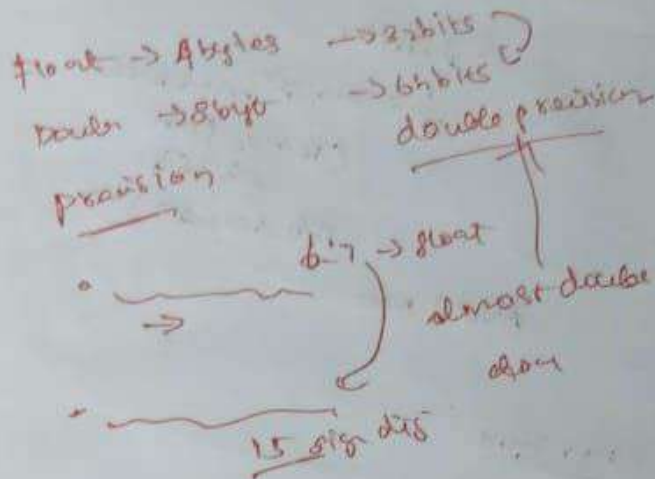
long  Billions Trillions  
 L script sometimes we may need move them Layer

Short Byte  Expensive (costly)  
 ↓  
More less things  
Costliest activities  
File name



## Float & Double

Double  $\rightarrow$  double precision  
same



Float  $\rightarrow$  Single Precision  $\rightarrow$  32 bits

Hexadecimal decimal

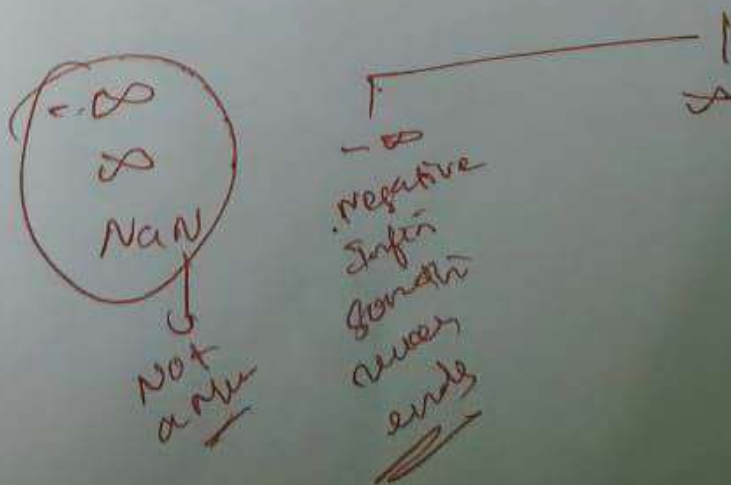
0x1.0113

Base 2 not 10

decimal

2.0

1.0 x 2<sup>-3</sup>



$$\begin{array}{r} 100 \\ 2 \overline{) 200} \\ \underline{200} \\ 0 \end{array}$$

$$\begin{array}{r} 0.38 \\ 3 \overline{) 109} \\ \underline{9} \\ 19 \end{array}$$

$$\frac{19}{38}$$

$$0.3333 \dots$$

Just like

$$\frac{1}{10} = 0.1 \text{ X not stored}$$

(Binary for 0.1)

No accurate

Avoid most

Char type

Think started of individual characters

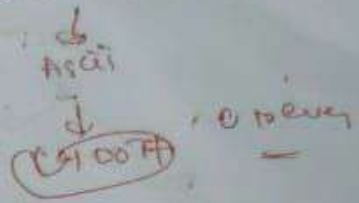
Unicode  $\rightarrow$  more than one char

Literally  $\rightarrow$  some place

U0000 to UFFFF

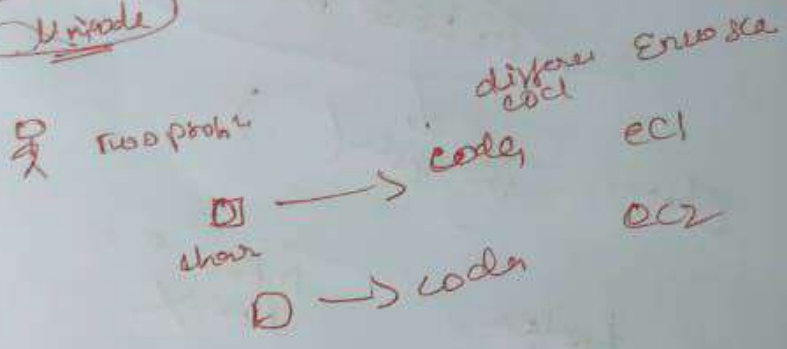
Escape Sequences  
 14 ->  
 others can print  
escape sequence

Unicode



Unicode  
 & processed  
 Before  
code is print

Unicode

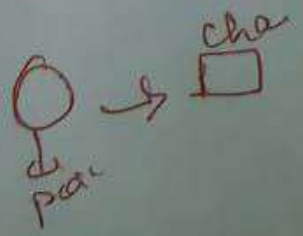


U -> two or more bits

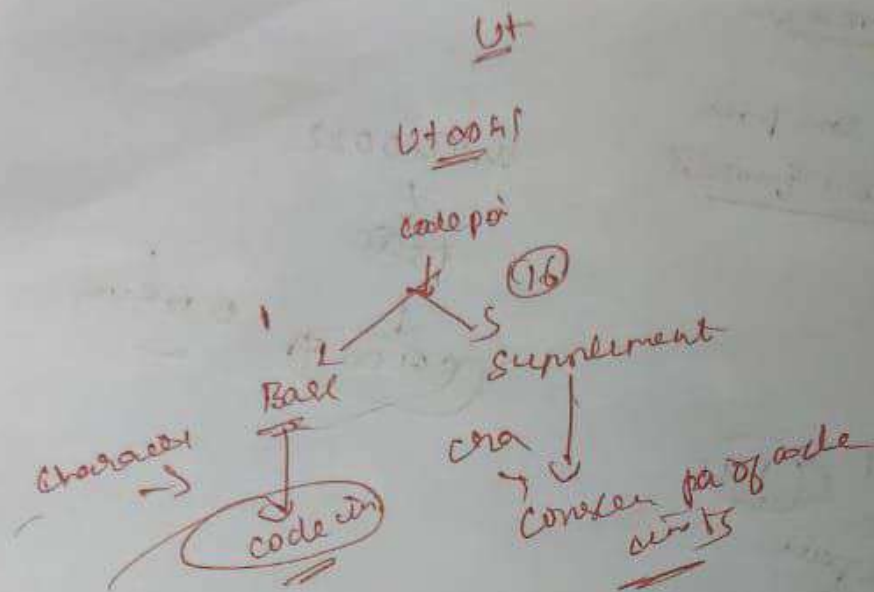
No unicode

Unicode 1991

code point -> code value  
 assigned with character







2048  
U+D800 to U+DBFF

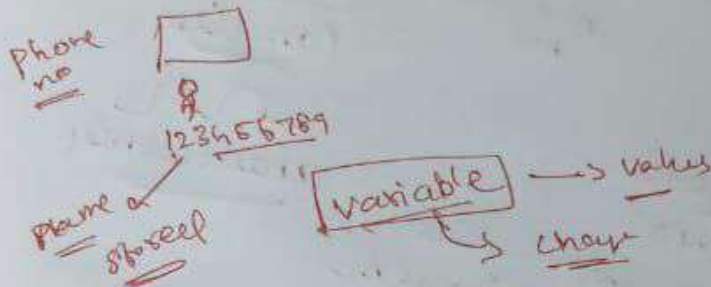
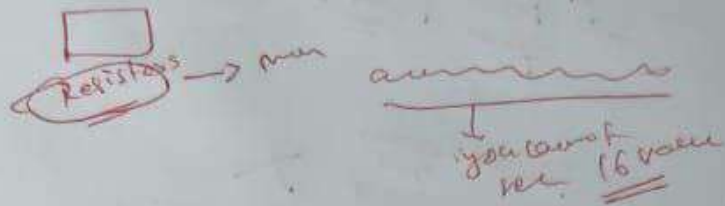
Character - Supplement

boolean

for pair

interior  
block

## Variables & Constants



### Constants

↓  
some one → constants  
unit  
char

### Declare

variable  
↓  
declare  
int x;  
int name of Boche;  
↓  
semicolon

### Rules

→ not start with digit  
letter, \_ , digit

for 0 X  
cancel

§ → valid  
§ into too names  
given by computer

→ reserve  
one  
reserved  
one

Camel Case

Initiation

int x;

abacus bottle

x = 10; → initialiser

§ local ones  
var

{

} → method block

const

Final → make it  
not can edit



## Enumerated

Sometimes you need  
particular set of var

pizzas { medium, small, large  
extra large

enum size {  
small  
MEDIUM  
LARGE  
}

size = small

}

## Operators

operator  
= Think of a man  
operates some  
machine

In simple Father work on light-bulb

Operations

screw  
driver



Tools → operators



→ operands

operands → values

operators → symbols

## Arithmetic

addition +  
 subtraction -  
 multiplication \*  
 division /  
 modulus %

$$00 + 0 = 000$$

$$00 - 0 = 00$$

$$00 * 00 \rightarrow 20$$

$$= 00$$

$$0000 = 2 * 00$$

$$00 \div 00 = 00$$

% seems left out

## Mathematical Functions & Const

Math . sqrt(x)

↓

↓

pow(x)

log10(x)

sin

cos

PI

float mod

$$n = 13$$

$$13 / 2 = 2$$

$$13 \% 2 = 1$$

$$13 / -2 = -2$$

$$13 \% -2 = -1$$

$$-13 \div 2 = 2$$

$$(-13 + 12) \div 2$$

↓  
shifts

$$-1 \div 2$$

$$(-1 \div 2 + 12) \div 2$$

→ shift

$$(-13 + 14) \div 2$$

$$1 \div 2$$

10 on hex



→ But for negative  
divisors

→ re val

8 bits math → 32 bits  
cells

~~NUMERIC TYPE~~  
conversion

Exact  
→ negative error  
abs error  
math